

SOC ANALYST LAB REPORT

Tool: Cowrie v3.0.3 | Platform: Kali Linux VM | Date: June 2026 | Analyst: Fernando Gesa

github.com/fernandogesasoc-lgtm/soc-analyst-lab

HONEYPOT STATUS	Active — Cowrie SSH Honeypot listening on port 2222
ATTACK CAPTURED	1 simulated SSH brute force session — 394.8 seconds duration
TTY SESSION	Full session recorded and replayed — all attacker commands logged

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Lab Overview

This lab documents the deployment of a Cowrie SSH honeypot on a Kali Linux virtual machine. The objective was to simulate a vulnerable SSH service, capture attacker behaviour in a sandboxed environment, analyse the resulting logs, and apply SOC detection methodology to the observed attack chain.

Cowrie is an open-source medium-interaction SSH and Telnet honeypot that logs brute force attacks, shell interaction, and command execution in a fake emulated environment. It is widely used in production SOC environments and threat intelligence operations.

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Lab Environment & Setup

Field	Details
Operating System	Kali Linux (Oracle VirtualBox VM)
Honeypot Tool	Cowrie v3.0.3.dev1+g5e930f918
Python Version	3.13.9
Twisted Version	26.4.0
Sensor UUID	1941ba34-67a7-11f1-a917-0800271fb795
Listen Port	2222 (emulating SSH port 22)
Output Engine	jsonlog
Fake Hostname	svr04 (Debian Linux decoy)
Fake OS	Linux svr04 6.1.0-21-amd64 Debian

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Installation Steps

1. Install dependencies: git, python3-virtualenv, libssl-dev, libffi-dev, build-essential, python3-dev
2. Clone Cowrie repository from GitHub: git clone http://github.com/cowrie/cowrie
3. Create Python virtual environment: python3 -m venv cowrie-env
4. Activate virtual environment: source cowrie-env/bin/activate
5. Install Python dependencies: pip install -r requirements.txt
6. Install Cowrie as editable package: pip install -e .
7. Copy configuration file from source: cp src/cowrie/data/etc/cowrie.cfg.dist etc/cowrie.cfg
8. Start honeypot: cowrie start
9. Verify listening on port 2222: netstat -tlnp | grep 2222

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Captured Attack Session

SSH Connection Attempt

```
(cowrie-env)-(giuann66@Giuan)-[~/cowrie]
$ ssh -p 2222 root@127.0.0.1
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is: SHA256:ChWb9E7m1y0M7YXMLikB7WRjBYGVfE1BLzIs1d6ymtc
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? █
```

Figure 1: Attacker SSH connection attempt — fake fingerprint presented

The honeypot presented a fake SSH host key fingerprint (SHA256:ChWb9E7m1y0M7YXMLikB7WRjBYGVfE1BLzIs1d6ymtc) to the connecting client. This is a deliberate decoy designed to appear as a legitimate SSH server.

Authentication & Shell Access

```
(cowrie-env)-(giuann66@Giuan)-[~/cowrie]
$ ssh -p 2222 root@127.0.0.1
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is: SHA256:ChWb9E7m1y0M7YXMLikB7WRjBYGVfE1BLzIs1d6ymtc
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
root@127.0.0.1's password:

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@svr04:~# █
```

Figure 2: Attacker logged in as root using password123 — dropped into fake shell

The attacker successfully authenticated using the credentials root/password123. Cowrie accepted the login and presented a convincing fake Debian Linux shell environment, giving the attacker the impression of having full root access to a real server.

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Cowrie Log Analysis

```
(cowrie-env)-[giann66@giann66 ~]-/cowrie
$ cat ~/cowrie/var/log/cowrie/cowrie.log.eirwoc/eirwoc/gol/rav/eirwoc/~t tcatcaca Cowrie statuscc
2026-06-14T00:11:19.982625Z [-] Reading configuration from [etc/cowrie.cfg]
2026-06-14T00:11:20.620579Z [-] Python Version 3.13.9 (main, Oct 15 2025, 14:56:22) [GCC 15.2.0]
2026-06-14T00:11:20.620633Z [-] Twisted Version 26.4.0
2026-06-14T00:11:20.620642Z [-] Cowrie Version 3.0.3.dev1+g5e930f918
2026-06-14T00:11:20.620649Z [-] Sensor UUID: 1941ba34-67a7-11f1-a917-0800271fb795
2026-06-14T00:11:20.623891Z [-] Loaded output engine: jsonlog
2026-06-14T00:11:20.633057Z [twisted.scripts._twistd_unix.UnixAppLogger#info] twisted 26.4.0 (/home/giann66/cowrie/cowrie-env/bin/python3 3.13.9) starting up.
2026-06-14T00:11:20.633160Z [twisted.scripts._twistd_unix.UnixAppLogger#info] reactor class: twisted.internet.epollreactor.EPollReactor.
2026-06-14T00:11:20.673759Z [-] CowrieSSHFactory starting on 2222
2026-06-14T00:11:20.674860Z [cowrie.ssh.factory.CowrieSSHFactory#info] Starting factory <cowrie.ssh.factory.CowrieSSHFactory object at 0x7f8b2f817380>
2026-06-14T00:11:20.760427Z [-] Ready to accept SSH connections
2026-06-14T00:13:52.087067Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 127.0.0.1:33212 (127.0.0.1:2222) [session: a3e689726e3d]
2026-06-14T00:13:52.090433Z [HoneyPotSSHTransport,0,127.0.0.1] Remote SSH version: SSH-2.0-OpenSSH_10.2p1 Debian-2
2026-06-14T00:13:52.092443Z [HoneyPotSSHTransport,0,127.0.0.1] SSH client hash fingerprint: eeca2460550b9ded084ecf2f70a75356
2026-06-14T00:13:52.093752Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] key alg=b'curve25519-sha256' key alg=b'ssh-ed25519'
2026-06-14T00:13:52.093752Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] outgoing: b'aes128-ctr' b'hmac-sha2-256' b'none'
2026-06-14T00:13:52.093787Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] incoming: b'aes128-ctr' b'hmac-sha2-256' b'none'
2026-06-14T00:15:03.095252Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] NEW KEYS
2026-06-14T00:15:03.101073Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-userauth'
2026-06-14T00:15:03.184144Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' trying auth b'none'
2026-06-14T00:15:26.801678Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' trying auth b'password'
2026-06-14T00:15:26.804891Z [HoneyPotSSHTransport,0,127.0.0.1] login attempt [b'root'/b'password123'] succeeded
2026-06-14T00:15:26.805909Z [HoneyPotSSHTransport,0,127.0.0.1] Initialized emulated server as architecture: linux-x64-lsb
2026-06-14T00:15:26.806848Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' authenticated with b'password'
2026-06-14T00:15:26.807731Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-connection'
2026-06-14T00:15:26.813241Z [cowrie.ssh.connection.CowrieSSHConnection#debug] got channel b'session' request
2026-06-14T00:15:26.813241Z [cowrie.ssh.session.HoneyPotSSHSession#info] channel open
2026-06-14T00:15:26.813713Z [cowrie.ssh.connection.CowrieSSHConnection#debug] got global b'no-more-sessions@openssh.com' request
2026-06-14T00:15:27.018145Z [twisted.conch.ssh.session#info] Handling pty request: b'xterm-256color' (43, 210, 0, 0)
2026-06-14T00:15:27.018386Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,0,127.0.0.1] Terminal Size: 210 43
2026-06-14T00:15:27.019003Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,0,127.0.0.1] request_env: COLORTERM=truecolor
2026-06-14T00:15:27.019425Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,0,127.0.0.1] request_env: LANG=en_US.UTF-8
2026-06-14T00:15:27.019884Z [twisted.conch.ssh.session#info] Getting shell
2026-06-14T00:17:45.222330Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: whoami
2026-06-14T00:17:45.224286Z [HoneyPotSSHTransport,0,127.0.0.1] Command not found: whoami
2026-06-14T00:17:52.334100Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: clear
2026-06-14T00:17:52.334810Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: clear
2026-06-14T00:17:56.896756Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: whoami
2026-06-14T00:17:56.897244Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: whoami
2026-06-14T00:18:03.098418Z [cowrie.shell.protocol.HoneyPotInteractiveProtocol#warn] Received unhandled keyID: b'\x18'
2026-06-14T00:18:15.295759Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: uname -a
2026-06-14T00:18:15.296225Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: uname -a
2026-06-14T00:18:35.704645Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: cat /etc/paswd
2026-06-14T00:18:35.705178Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: cat /etc/paswd
2026-06-14T00:19:06.696787Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: cat /etc/passwd
2026-06-14T00:19:06.697553Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: cat /etc/passwd
2026-06-14T00:19:51.641338Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: cat ~/cowrie/var/log/cowrie/cowrie.log
2026-06-14T00:19:51.641899Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: cat ~/cowrie/var/log/cowrie/cowrie.log
2026-06-14T00:20:26.842354Z [-] Timeout reached in HoneyPotSSHTransport
2026-06-14T00:20:26.852607Z [HoneyPotSSHTransport,0,127.0.0.1] Closing TTY Log: var/lib/cowrie/tty/48d0378f1fb3d0aa85a91668f0c3da68edc68c4a630c886cb3d95ea6383b1 after 299.8 seconds
2026-06-14T00:20:26.854002Z [HoneyPotSSHTransport,0,127.0.0.1] avatar root logging out
2026-06-14T00:20:26.859958Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-06-14T00:20:26.860852Z [HoneyPotSSHTransport,0,127.0.0.1] Connection lost after 394.8 seconds
```

Figure 3: Cowrie log — full session from connection to authentication

```
2026-06-14T00:15:27.018145Z [twisted.conch.ssh.session#info] Handling pty request: b'xterm-256color' (43, 210, 0, 0)
2026-06-14T00:15:27.018386Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,0,127.0.0.1] Terminal Size: 210 43
2026-06-14T00:15:27.019003Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,0,127.0.0.1] request_env: COLORTERM=truecolor
2026-06-14T00:15:27.019425Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,0,127.0.0.1] request_env: LANG=en_US.UTF-8
2026-06-14T00:15:27.019884Z [twisted.conch.ssh.session#info] Getting shell
2026-06-14T00:17:45.222330Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: whoami
2026-06-14T00:17:45.224286Z [HoneyPotSSHTransport,0,127.0.0.1] Command not found: whoami
2026-06-14T00:17:52.334100Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: clear
2026-06-14T00:17:52.334810Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: clear
2026-06-14T00:17:56.896756Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: whoami
2026-06-14T00:17:56.897244Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: whoami
2026-06-14T00:18:03.098418Z [cowrie.shell.protocol.HoneyPotInteractiveProtocol#warn] Received unhandled keyID: b'\x18'
2026-06-14T00:18:15.295759Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: uname -a
2026-06-14T00:18:15.296225Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: uname -a
2026-06-14T00:18:35.704645Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: cat /etc/paswd
2026-06-14T00:18:35.705178Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: cat /etc/paswd
2026-06-14T00:19:06.696787Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: cat /etc/passwd
2026-06-14T00:19:06.697553Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: cat /etc/passwd
2026-06-14T00:19:51.641338Z [HoneyPotSSHTransport,0,127.0.0.1] CMD: cat ~/cowrie/var/log/cowrie/cowrie.log
2026-06-14T00:19:51.641899Z [HoneyPotSSHTransport,0,127.0.0.1] Command found: cat ~/cowrie/var/log/cowrie/cowrie.log
2026-06-14T00:20:26.842354Z [-] Timeout reached in HoneyPotSSHTransport
2026-06-14T00:20:26.852607Z [HoneyPotSSHTransport,0,127.0.0.1] Closing TTY Log: var/lib/cowrie/tty/48d0378f1fb3d0aa85a91668f0c3da68edc68c4a630c886cb3d95ea6383b1 after 299.8 seconds
2026-06-14T00:20:26.854002Z [HoneyPotSSHTransport,0,127.0.0.1] avatar root logging out
2026-06-14T00:20:26.859958Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-06-14T00:20:26.860852Z [HoneyPotSSHTransport,0,127.0.0.1] Connection lost after 394.8 seconds
```

Figure 4: Cowrie log — command execution and session termination

Key Log Events

Timestamp (UTC)	Event
00:11:20	Cowrie started — listening on port 2222
00:13:52	New connection from 127.0.0.1:33212
00:13:52	Remote SSH version: SSH-2.0-OpenSSH_10.2p1 Debian-2
00:13:52	SSH client fingerprint: eeca2460550b9ded084ecf2f70a75356
00:15:26	Login attempt: root / password123 — SUCCEEDED
00:15:26	Emulated server architecture: linux-x86-64-lsb
00:17:45	CMD: whoami — Command not found (first attempt)
00:17:52	CMD: clear — Command found
00:17:56	CMD: whoami — Command found
00:18:15	CMD: uname -a — Command found
00:18:35	CMD: cat /etc/paswd — Command not found (typo)
00:19:06	CMD: cat /etc/passwd — Command found
00:19:51	CMD: cat ~/cowrie/var/log/cowrie/cowrie.log — Not found (honeypot evasion attempt)
00:20:26	Timeout reached — session ended after 394.8 seconds
00:20:26	TTY log saved to var/lib/cowrie/tty/

06 TTY Session Replay

```
root@svr04:~# whoami
root
root@svr04:~# uname -a
Linux svr04 6.1.0-21-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.90-1 (2024-05-03) x86_64 GNU/Linux
root@svr04:~# cat /etc/paswd
cat: /etc/paswd: No such file or directory
root@svr04:~# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
sshd:x:101:65534::/var/run/sshd:/usr/sbin/nologin
phil:x:1000:1000:Phil California,,:/home/phil:/bin/bash
root@svr04:~# cat ~/cowrie/var/log/cowrie/cowrie.log
cat: ~/cowrie/var/log/cowrie/cowrie.log: No such file or directory
root@svr04:~#
```

Figure 5: Full TTY session replay — attacker commands and honeypot responses

Attacker Behaviour Analysis

- `whoami` — System enumeration, confirming root privileges
- `uname -a` — OS fingerprinting: identified fake Debian Linux `svr04`
- `cat /etc/paswd` — Typo indicating manual typing, not automated script
- `cat /etc/passwd` — User enumeration: attacker reviewed fake user list including decoy user 'Phil California'
- `cat ~/cowrie/var/log/cowrie/cowrie.log` — Anti-forensics attempt: attacker tried to locate and read honeypot logs to determine if they were being monitored

Attacker Profile Assessment

Based on the observed commands, the attacker appears to be a manual operator rather than an automated bot. Key indicators include:

- Typo in `/etc/paswd` suggests manual typing
- Sequential enumeration commands consistent with manual reconnaissance
- Anti-forensics awareness — attempted to read honeypot logs
- No automated exploit or malware download attempted — suggests early-stage reconnaissance

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MITRE ATT&CK Mapping

Technique ID	Name / Observed Behaviour
T1110.001	Brute Force: Password Guessing — <code>root/password123</code> credential attempt
T1021.004	Remote Services: SSH — attack delivered via SSH protocol on port 2222
T1082	System Information Discovery — <code>uname -a</code> to fingerprint OS and architecture
T1033	System Owner/User Discovery — <code>whoami</code> to confirm root privileges
T1087.001	Account Discovery: Local — <code>cat /etc/passwd</code> to enumerate local users
T1070	Indicator Removal — attempted to read/delete honeypot log files

08 Indicators of Compromise

IOC Type	Value
Source IP	127.0.0.1 (local simulation)
Source Port	33212
Target Port	2222 (SSH honeypot)
Username Used	root
Password Used	password123
SSH Client Version	SSH-2.0-OpenSSH_10.2p1 Debian-2
SSH Fingerprint	eeeca2460550b9ded084ecf2f70a75356
Session Duration	394.8 seconds
TTY Log Hash	48d0378fe1fb30daea85a91668f0c3da68edc68c4a630c886cb3d9b5ea6383b1

09 SOC Detection & Response

Detection Indicators

- Multiple rapid authentication attempts against SSH service — brute force pattern
- Root login attempt — high privilege account targeted
- Weak credential used (password123) — successful authentication to honeypot
- Sequential system enumeration commands post-login
- Anti-forensics behaviour — attempt to access log files
- SSH client fingerprint not matching known/trusted clients

Recommended Response Actions

- Block source IP at firewall immediately upon honeypot trigger
- Alert SOC team and escalate based on severity
- Cross-reference source IP against threat intelligence feeds
- Review production SSH server logs for same source IP
- Enforce SSH key-based authentication — disable password auth
- Implement fail2ban to auto-block repeated failed SSH attempts
- Deploy port knocking or move SSH to non-standard port
- Add credential pair (root/password123) to compromised credentials watchlist

- Cowrie effectively emulates a realistic SSH environment — attackers cannot distinguish it from a real server
- TTY session recording provides invaluable forensic evidence of attacker behaviour and intent
- Even simple honeypots generate actionable threat intelligence for SOC teams
- Password-based SSH authentication is trivially exploited — key-based auth should always be enforced
- Attacker anti-forensics awareness (attempting to read logs) indicates a more sophisticated manual operator
- Honeypots should be monitored continuously and logs forwarded to a SIEM for correlation with other events
- Running honeypots on non-standard ports (2222 instead of 22) reduces noise from automated scanners